

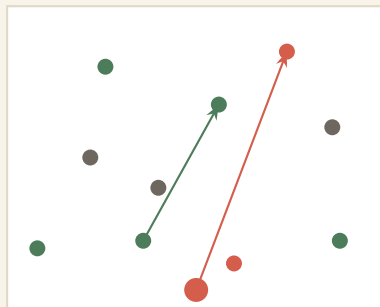
Effects of Superspreaders in Spread of Epidemic

Spatial SIR model, Monte Carlo simulation, and SARS comparison

Reference: Fujie and Odagaki, *Physica A*, 2007

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local and long-range
transmission paths

Roadmap

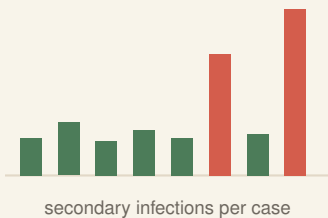
1. Problem and modeling approach
2. Spatial SIR model with two superspreader mechanisms
3. Monte Carlo solving algorithm
4. Reproduced simulation results
5. Sensitivity, robustness, and reference conclusion

Problem

- ▶ Classical well-mixed SIR models assume similar transmission rates for all infected individuals.
- ▶ Real outbreaks often show strong heterogeneity: a small number of cases cause many secondary infections.
- ▶ The paper asks whether epidemic spread is driven more by **strong local infectiousness** or by **high connectivity**.

Key question

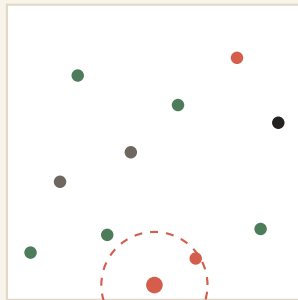
Are superspreaders important because they are more infectious locally, or because they have many social connections?



Modeling Approach

- ▶ Individuals are fixed in a two-dimensional continuous domain of side length $L = 10r_0$.
- ▶ Each individual has state S , I , or R :
 $X_i(t) \in \{S, I, R\}$.
- ▶ Transmission probability depends on periodic distance r between an infected individual and a susceptible individual.
- ▶ A fraction λ of the population is assigned as superspreaders.

$$\rho = \frac{N}{L^2}.$$



periodic boundary

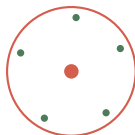
initial infected at bottom

Two Superspreader Models

Strong infectiousness

Same local radius, higher infection probability.

$$w_s(r) = \begin{cases} w_0, & 0 \leq r \leq r_0, \\ 0, & r > r_0. \end{cases}$$



same range r_0

Hub model

Same probability shape, wider contact range.

$$w_s(r) = \begin{cases} w_0(1 - r/r_n)^2, & 0 \leq r \leq r_n, \\ 0, & r > r_n, \end{cases}$$

$$r_n = \sqrt{6}r_0.$$

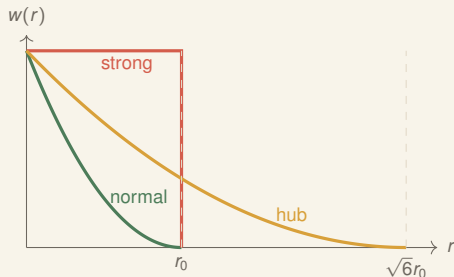
Long-range contacts change spatial spread, even with comparable expected infection strength.

Normal vs Superspreader Infection Kernels

Normal infected individual

$$w_n(r) = \begin{cases} w_0(1 - r/r_0)^2, & 0 \leq r \leq r_0, \\ 0, & r > r_0. \end{cases}$$

- ▶ Local transmission.
- ▶ Probability decays with distance.



Normalization idea

The hub radius $\sqrt{6}r_0$ makes the expected infection strength comparable to the strong infectiousness model.

Basic Reproductive Number and Critical Density

$R_0(\lambda)$ links microscopic infection kernels to the density threshold.

$$R_0(\lambda) = \frac{\rho}{w_0} \left[\lambda \int w_s(r) 2\pi r dr + (1 - \lambda) \int w_n(r) 2\pi r dr \right].$$

$$R_0 = R_c, \quad \rho_c \pi r_0^2 = \frac{R_c}{\lambda + \frac{1-\lambda}{6}}.$$

Strong infectiousness

$$R_c \approx 4.5$$

Hub model

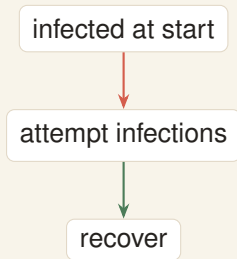
$$R_c \approx 3.2$$

Monte Carlo Solving Algorithm

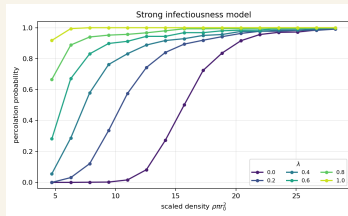
1. Randomly place N individuals; set one initial infected individual.
2. Randomly assign superspreader labels using fraction λ .
3. Infected-at-start individuals attempt to infect susceptible individuals with probability $w(r)$.
4. Newly infected individuals transmit only from the next step.
5. Current infected individuals recover with probability γ .
6. Stop when no infected individuals remain or the maximum time is reached.

Output quantities

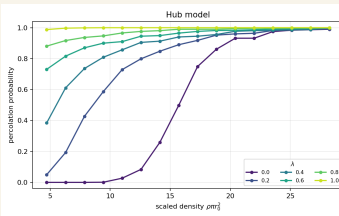
- ▶ percolation probability
- ▶ propagation front $r_f(t)$
- ▶ epidemic curve
- ▶ secondary infection distribution



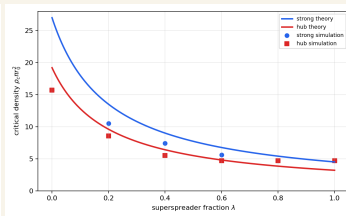
Result 1: Percolation



Strong



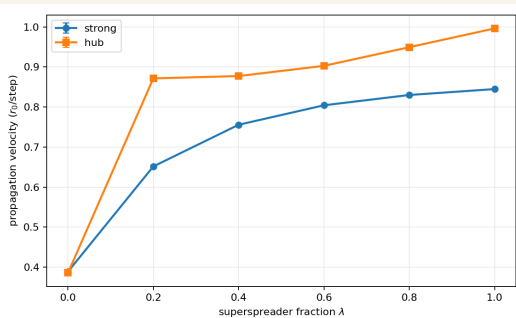
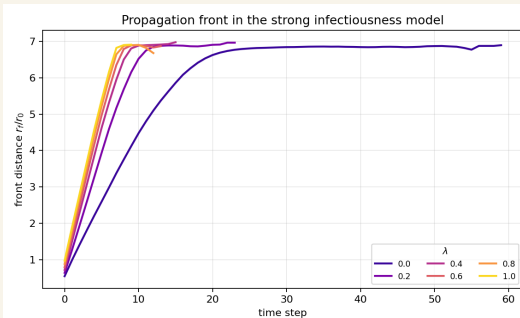
Hub



Critical density

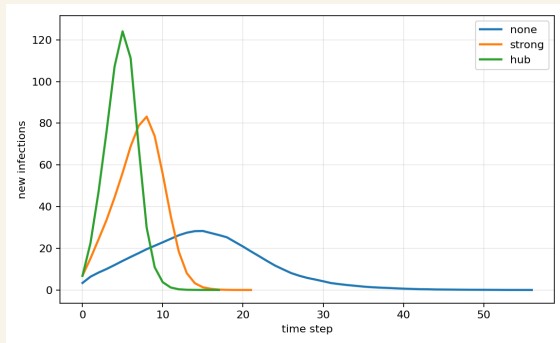
Increasing λ lowers the density threshold for percolation. The hub model percolates at lower density.

Result 2: Propagation Speed



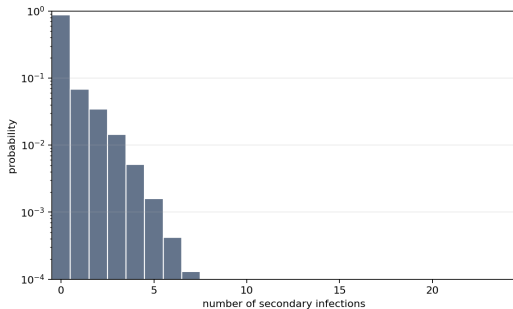
Propagation speed increases with λ . For $\lambda > 0$, the hub model spreads faster because it creates long-range infection paths.

Result 3: Epidemic Curves

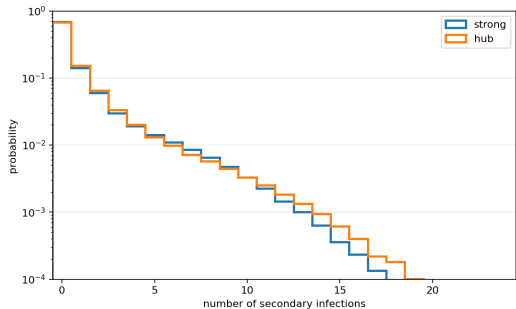


- ▶ No-superspreader outbreaks are broader and peak later.
- ▶ Superspreaders make the peak earlier and sharper.
- ▶ The hub model produces the fastest rise.

Result 4: Secondary Infections



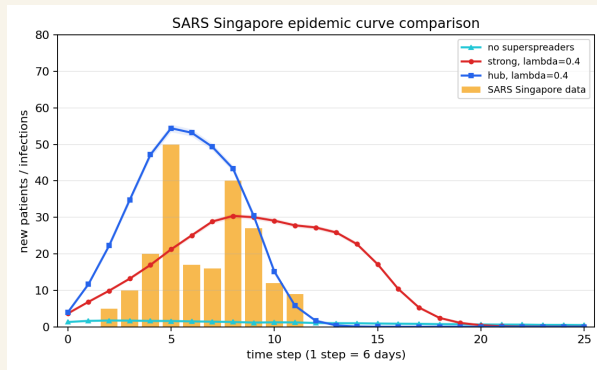
No superspreaders



With superspreaders, $\lambda = 0.2$

Superspreaders create a long-tailed secondary-infection distribution, matching the qualitative pattern discussed in the paper.

Comparison with SARS Data



- ▶ The reference paper compares the model with SARS data from Singapore in 2003.
- ▶ The hub model with $N = 477$, $\rho\pi r_0^2 = 15.0$, $\lambda = 0.4$, and $\gamma = 1$ gives the best qualitative epidemic-curve match. SARS superspreading was better explained by many social connections than by stronger local infectiousness alone.

Sensitivity and Robustness

- ▶ The model is sensitive to superspreader fraction λ : increasing λ lowers critical density and increases propagation speed.
- ▶ It is sensitive to the mechanism of superspreading: hub contacts produce faster spatial spread than stronger local infectiousness.
- ▶ Population density controls whether infection chains die out or percolate.
- ▶ Recovery probability γ is also important: lower γ means a longer infectious period and more transmission opportunities.

Robust qualitative findings

Across parameter settings, superspreaders lower the percolation threshold, speed up spread, sharpen epidemic curves, and create long-tailed secondary-infection distributions.

Limitations

- ▶ Individuals are fixed in space, while real people move and change contacts.
- ▶ Superspreader labels are assigned randomly, but real superspreading depends on behavior, occupation, environment, and biology.
- ▶ The model is best interpreted as a controlled comparison of two mechanisms, not a full prediction tool for every epidemic.

Conclusion

1. Superspreaders substantially change epidemic dynamics.
2. The hub model spreads faster and has lower critical density than the strong infectiousness model.
3. Long-range social connectivity is the key mechanism in the reference paper's SARS comparison.

Takeaway

Average transmission rate alone is not enough; epidemic models should include heterogeneity in individual connectivity.